

22	B	Since the heating elements are in parallel, they have the same p.d. across them. $\text{Power } P = \frac{V^2}{R} = \frac{V^2}{\frac{\rho L}{A}} = \frac{AV^2}{\rho L} = \frac{\pi \left(\frac{d}{2}\right)^2 V^2}{\rho L}$ $P \propto d^2$ since p.d., resistivity and length are the same. $\frac{d_x}{d_y} = \left(\frac{P_x}{P_y}\right)^{0.5} = \left(\frac{1.0}{1.5}\right)^{0.5} = 0.82$
23	A	$I = 3I$